

PrimeCore Minerals

HQ Network

Monitoring and

Logging

Implementation

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Introduction

This document provides an explanation of how the network monitoring and centralised logging for the PrimeCore Minerals Brisbane HQ were set up. The HQ-MONITOR server is used to collect and view network activity and pfSense security events. SPAN is used to capture network traffic, while pfSense logs are sent to the server and displayed through a Grafana dashboard.

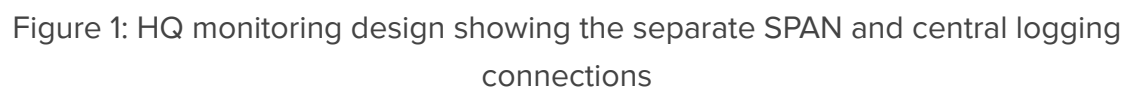
Purpose

Monitoring and logging are part of the project MVP and are required to provide visibility into the activity taking place across the PrimeCore network.

By implementing the monitoring and logging setup, it progresses the project towards the MVP by giving the team a clear way to monitor activity across the Brisbane HQ network. The monitoring solutions that have been is an important component for my role as Security and Validation Lead, as they allow me to verify network traffic and security events during testing and ensure security controls are functioning as intended.

SPAN monitoring lets me view copied network traffic in a simple and easy way, while pfSense logging and Grafana provide a visual representation of firewall traffic and events.

The HQ-MONITOR server was added to Server VLAN 90 and uses two network interfaces for different monitoring tasks. I configured SPAN monitoring through Open vSwitch to copy network traffic to the monitoring server. SPAN works by copying traffic from a selected source and sending that copy to another interface for monitoring. The pfSense trunk port ens4 on the Open vSwitch was used as the source because it carries traffic between pfSense and the different VLANs.



Interface	Purpose	Configuration
ens3	Central logging and Grafana access	192.168.90.60/24
ens4	SPAN packet monitoring	None

SPAN Network Monitoring

To make the activity on the Brisbane HQ network more accessible for viewing, I created a monitoring server and configured SPAN monitoring through Open vSwitch on it. SPAN works by creating a copy of traffic from a specific source and sending it to a specified device. The pfSense trunk interface 'ens4' was used as the source because it is the main connection carrying network traffic between pfSense and the different VLANs. The traffic then gets copied to ens12, which is the mirror output connected to HQ-MONITOR.

```
sudo ovs-vsctl \  
-- --id=@src get Port ens4 \  
-- --id=@dst get Port ens12 \  
-- --id=@m create Mirror name=HQ-SPAN \  
select-src-port=@src \  
select-dst-port=@src \  
output-port=@dst \  
-- set Bridge br0 mirrors=@m
```

```
sudo tcpdump -eni ens4
```

The packet capture showed traffic from multiple VLANs and displayed the VLAN tags associated with the packets.

```

length 0
14:55:52.962292 50:00:00:01:00:02 > 00:50:00:00:07:00, ethertype 802.1Q (0x8100), length 101: vlan 10, p 0, ethertype IPv4 (0x0800), 104.20.23.154.443 > 192.168.10.72.43584: Flags [P.], seq 5624:5655, ack 1828, win 16, options [nop,nop,TS val 4196960388 ecr 3767957144], length 31
14:55:52.962431 00:50:00:00:07:00 > 50:00:00:01:00:02, ethertype 802.1Q (0x8100), length 101: vlan 10, p 0, ethertype IPv4 (0x0800), 192.168.10.72.43584 > 104.20.23.154.443: Flags [P.], seq 1828:1859, ack 5624, win 302, options [nop,nop,TS val 3767957148 ecr 4196960387], length 31
14:55:52.968093 50:00:00:01:00:02 > 00:50:00:00:07:00, ethertype 802.1Q (0x8100), length 229: vlan 10, p 0, ethertype IPv4 (0x0800), 104.20.23.154.443 > 192.168.10.72.43584: Flags [P.], seq 5655:5814, ack 1859, win 16, options [nop,nop,TS val 4196960394 ecr 3767957148], length 159
14:55:52.968566 00:50:00:00:07:00 > 50:00:00:01:00:02, ethertype 802.1Q (0x8100), length 70: vlan 10, p 0, ethertype IPv4 (0x0800), 192.168.10.72.43584 > 104.20.23.154.443: Flags [I.], ack 5814, win 314, options [nop,nop,TS val 3767957154 ecr 4196960388], length 0
14:55:52.970869 00:50:00:00:07:00 > 50:00:00:01:00:02, ethertype 802.1Q (0x8100), length 118: vlan 10, p 0, ethertype IPv4 (0x0800), 192.168.10.72.43584 > 104.20.23.154.443: Flags [P.], seq 1859:1907, ack 5814, win 314, options [nop,nop,TS val 3767957156 ecr 4196960388], length 48
14:55:52.970870 00:50:00:00:07:00 > 50:00:00:01:00:02, ethertype 802.1Q (0x8100), length 94: vlan 10, p 0, ethertype IPv4 (0x0800), 192.168.10.72.43584 > 104.20.23.154.443: Flags [P.], seq 1907:1931, ack 5814, win 314, options [nop,nop,TS val 3767957157 ecr 4196960388], length 24
14:55:52.971990 00:50:00:00:07:00 > 50:00:00:01:00:02, ethertype 802.1Q (0x8100), length 70: vlan 10, p 0, ethertype IPv4 (0x0800), 192.168.10.72.43584 > 104.20.23.154.443: Flags [F.], seq 1931, ack 5814, win 314, options [nop,nop,TS val 3767957157 ecr 4196960388], length 0
14:55:52.975273 50:00:00:01:00:02 > 00:50:00:00:07:00, ethertype 802.1Q (0x8100), length 70: vlan 10, p 0, ethertype IPv4 (0x0800), 104.20.23.154.443 > 192.168.10.72.43584: Flags [I.], ack 1931, win 16, options [nop,nop,TS val 4196960401 ecr 3767957156], length 0
14:55:52.976626 50:00:00:01:00:02 > 00:50:00:00:07:00, ethertype 802.1Q (0x8100), length 70: vlan 10, p 0, ethertype IPv4 (0x0800), 104.20.23.154.443 > 192.168.10.72.43584: Flags [F.], seq 5814, ack 1932, win 16, options [nop,nop,TS val 4196960402 ecr 3767957157], length 0
14:55:52.977029 00:50:00:00:07:00 > 50:00:00:01:00:02, ethertype 802.1Q (0x8100), length 70: vlan 10, p 0, ethertype IPv4 (0x0800), 192.168.10.72.43584 > 104.20.23.154.443: Flags [I.], ack 5815, win 314, options [nop,nop,TS val 3767957163 ecr 4196960402], length 0
14:55:57.582489 00:50:00:00:0b:00 > ff:ff:ff:ff:ff:ff, ethertype 802.1Q (0x8100), length 331: vlan 90, p 0, ethertype IPv4 (0x0800), 0.0.0.0.68 > 255.255.255.255.67: BOOTP/DHCP, Request from 00:50:00:00:0b:00, length 285
14:55:58.702738 00:50:00:00:0b:01 > ff:ff:ff:ff:ff:ff, ethertype IPv4 (0x0800), length 327: 0.0.0.0.68 > 255.255.255.255.67: BOOTP/DHCP, Request from 00:50:00:00:0b:01, length 285
14:56:40.509848 00:50:00:00:08:00 > 50:00:00:01:00:02, ethertype 802.1Q (0x8100), length 102: vlan 20, p 0, ethertype IPv4 (0x0800), 192.168.20.72 > 192.168.10.72: ICMP echo request, id 19332, seq 1, length 64
14:56:41.557943 00:50:00:00:08:00 > 50:00:00:01:00:02, ethertype 802.1Q (0x8100), length 102: vlan 20, p 0, ethertype IPv4 (0x0800), 192.168.20.72 > 192.168.10.72: ICMP echo request, id 19332, seq 2, length 64
14:56:42.581972 00:50:00:00:08:00 > 50:00:00:01:00:02, ethertype 802.1Q (0x8100), length 102: vlan 20, p 0, ethertype IPv4 (0x0800), 192.168.20.72 > 192.168.10.72: ICMP echo request, id 19332, seq 3, length 64
14:56:43.606040 00:50:00:00:08:00 > 50:00:00:01:00:02, ethertype 802.1Q (0x8100), length 102: vlan 20, p 0, ethertype IPv4 (0x0800), 192.168.20.72 > 192.168.10.72: ICMP echo request, id 19332, seq 4, length 64
14:56:45.910186 00:50:00:00:08:00 > 50:00:00:01:00:02, ethertype 802.1Q (0x8100), length 46: vlan 20, p 0, ethertype ARP (0x0806), Request who-has 192.168.20.1 tell 192.168.20.72, length 28
14:56:45.910861 50:00:00:01:00:02 > 00:50:00:00:08:00, ethertype 802.1Q (0x8100), length 46: vlan 20, p 0, ethertype ARP (0x0806), Reply 192.168.20.1 is-at 50:00:00:01:00:02, length 28

```

Figure 2: SPAN packet capture on HQ-MONITOR displaying copied traffic from trunk.

Centralised pfSense Logging

In addition to SPAN monitoring, pfSense was configured to send security and other system events to the HQ-MONITOR

The logs are sent through VLAN 90 on ens3 and HQ-MONITOR, which uses the static IP address 192.168.90.60/24.

Using syslogs over UDP port 514, pfSense sends the events to the monitoring server. rsyslog on HQ-MONITOR receives the events and stores them in /var/log/pfsense.log.

The dedicated rsyslog configuration was created on HQ-MONITOR using:

```
sudo nano /etc/rsyslog.d/10-pfsense.conf
```

The rsyslog configuration uses the block:

```
module(load="imudp")
input(type="imudp" port="514")

if ($fromhost-ip == '192.168.90.1') then {
    action(type="omfile" file="/var/log/pfsense.log")
    stop
}
```

This rsyslog configuration tells HQ-MONITOR to listen for syslog messages on UDP port 514. It verifies the messages are coming from pfSense at 192.168.90.1. If they are, the events are saved into /var/log/pfsense.log so the pfSense logs can be stored and reviewed from the monitoring server.

```

labadmin@srv-lnx-01:~$ sudo tail -f /var/log/pfsense.log
[sudo: authenticate] Password:
2026-09-11T06:09:15+00:00 192.168.90.1 filterlog[62179]: 1,71,,12004,vtnet0,match,block,in,4,0x0,,64,17412,0,DF,17,udp,250,192.1
68.1.169,192.168.1.255,6537,6537,230
2026-09-11T06:09:15+00:00 192.168.90.1 filterlog[62179]: 1,71,,12004,vtnet0,match,block,in,4,0x0,,64,16487,0,DF,17,udp,32,192.16
8.1.32,255.255.255.255,40706,10001,12
2026-09-11T06:09:15+00:00 192.168.90.1 filterlog[62179]: 1,71,,12004,vtnet0,match,block,in,4,0x0,,64,17424,0,DF,17,udp,250,192.1
68.1.169,192.168.1.255,6537,6537,230
2026-09-11T06:09:15+00:00 192.168.90.1 filterlog[62179]: 1,71,,12004,vtnet0,match,block,in,4,0x0,,64,6824,0,DF,17,udp,216,192.16
8.1.68,255.255.255.255,34206,6667,196
2026-09-11T06:09:16+00:00 192.168.90.1 filterlog[62179]: 1,71,,12004,vtnet0,match,block,in,4,0x0,,64,0,0,DF,17,udp,248,192.168.1
.221,255.255.255.255,54286,6667,228
2026-09-11T06:09:17+00:00 192.168.90.1 filterlog[62179]: 1,71,,12004,vtnet0,match,block,in,4,0x0,,255,34077,0,none,17,udp,216,19
2.168.1.27,255.255.255.255,49154,6667,196
2026-09-11T06:09:18+00:00 192.168.90.1 filterlog[62179]: 1,71,,12004,vtnet0,match,block,in,4,0x0,,255,27646,0,none,17,udp,216,19
2.168.1.26,255.255.255.255,49154,6667,196
2026-09-11T06:09:19+00:00 192.168.90.1 filterlog[62179]: 1,71,,12004,vtnet0,match,block,in,4,0x0,,255,29754,0,none,17,udp,216,19
2.168.1.28,255.255.255.255,49154,6667,196
2026-09-11T06:09:19+00:00 192.168.90.1 filterlog[62179]: 1,71,,12004,vtnet0,match,block,in,4,0x0,,255,1554,0,none,17,udp,200,192
.168.1.99,255.255.255.255,65060,6667,180
2026-09-11T06:09:19+00:00 192.168.90.1 filterlog[62179]: 1,71,,12004,vtnet0,match,block,in,4,0x0,,64,54280,0,DF,17,udp,200,192.1
68.1.229,255.255.255.255,49597,6667,180
2026-09-11T06:09:20+00:00 192.168.90.1 filterlog[62179]: 1,71,,12004,vtnet0,match,block,in,4,0x0,,64,17750,0,DF,17,udp,250,192.1
68.1.169,192.168.1.255,6537,6537,230
2026-09-11T06:09:20+00:00 192.168.90.1 filterlog[62179]: 1,71,,12004,vtnet0,match,block,in,4,0x0,,64,17773,0,DF,17,udp,250,192.1
68.1.169,192.168.1.255,6537,6537,230
2026-09-11T06:09:20+00:00 192.168.90.1 filterlog[62179]: 1,71,,12004,vtnet0,match,block,in,4,0x0,,64,6825,0,DF,17,udp,216,192.16
8.1.68,255.255.255.255,34206,6667,196
2026-09-11T06:09:21+00:00 192.168.90.1 filterlog[62179]: 1,71,,12004,vtnet0,match,block,in,4,0x0,,64,0,0,DF,17,udp,248,192.168.1
.221,255.255.255.255,54286,6667,228
2026-09-11T06:09:21+00:00 192.168.90.1 filterlog[62179]: 1,71,,12004,vtnet0,match,block,in,4,0x0,,255,34078,0,none,17,udp,216,19
2.168.1.27,255.255.255.255,49154,6667,196

```

Figure 3: The stored /var/log/pfsense.log pfSense events displayed on MQ-MONITOR

✗	Aug 26 13:39:13	WAN	192.168.1.229:39666	255.255.255.255:6667	UDP
✗	Aug 26 13:39:14	vtnet2	0.0.0.0:68	255.255.255.255:67	UDP
✗	Aug 26 13:39:14	WAN	192.168.1.99:65060	255.255.255.255:6667	UDP
✗	Aug 26 13:39:14	WAN	192.168.1.28:49154	255.255.255.255:6667	UDP
✗	Aug 26 13:39:15	WAN	192.168.1.221:54286	255.255.255.255:6667	UDP
✗	Aug 26 13:39:15	WAN	192.168.1.26:49154	255.255.255.255:6667	UDP
✗	Aug 26 13:39:15	EMPLOYEE_VLAN	0.0.0.0	192.168.10.1	ICMP
✗	Aug 26 13:39:15	WAN	192.168.1.32:57801	255.255.255.255:10001	UDP
✗	Aug 26 13:39:16	WAN	192.168.1.27:49154	255.255.255.255:6667	UDP
✗	Aug 26 13:39:17	EMPLOYEE_VLAN	0.0.0.0	192.168.10.1	ICMP
✗	Aug 26 13:39:18	WAN	192.168.1.229:39666	255.255.255.255:6667	UDP
✗	Aug 26 13:39:19	WAN	192.168.1.99:65060	255.255.255.255:6667	UDP
✗	Aug 26 13:39:19	WAN	192.168.1.28:49154	255.255.255.255:6667	UDP
✗	Aug 26 13:39:19	WAN	192.168.1.221:54286	255.255.255.255:6667	UDP
✗	Aug 26 13:39:20	WAN	192.168.1.26:49154	255.255.255.255:6667	UDP
✗	Aug 26 13:39:21	WAN	192.168.1.27:49154	255.255.255.255:6667	UDP
✗	Aug 26 13:39:23	WAN	192.168.1.229:39666	255.255.255.255:6667	UDP

Figure 4: pfSense firewall logs showing blocked network traffic.

Grafana Monitoring

Although the pfSense logs could already be viewed through the HQ-MONITOR terminal, it was not a permanent solution and was difficult to understand. To provide a more improved form of monitoring and logging, I installed Grafana, Loki and Grafana Alloy on HQ-MONITOR so the pfSense events could be viewed through a visual dashboard instead.

Tool	Function
Grafana Alloy	Reads the pfSense log file and forwards the events
Loki	Stores and searches the collected logs
rsyslog	Receives pfSense syslog events
Grafana	Displays the logs and other information through a custom web dashboard

To configure Grafana Alloy, I opened its configuration file using:

```
sudo nano /etc/alloy/config.alloy
```

I then added the following command:

```
loki.source.file "pfsense" {
  targets = [
    {
      __path__ = "/var/log/pfsense.log",
      job      = "pfsense",
      site     = "brisbane-hq",
    },
  ]

  forward_to = [loki.write.local.receiver]
}
loki.write "local" {
  endpoint {
    url = "http://127.0.0.1:3100/loki/api/v1/push"
```

```
}  
}
```

This configuration tells Grafana Alloy to read the pfSense logs within /var/log/pfsense.log and send them into Loki. The logs are then provided a label "brisbane-hq" so they can be displayed in Grafana.

Grafana then gets accessed from HQ-MGMT-PC using the URL <http://192.168.90.60:3000>. Within Grafana I added Loki as the data source using http://127.0.0.1:3100. I used this address because Grafana and Loki are both running on the same HQ-MONITOR server, which allows Grafana to access the Loki logs locally.

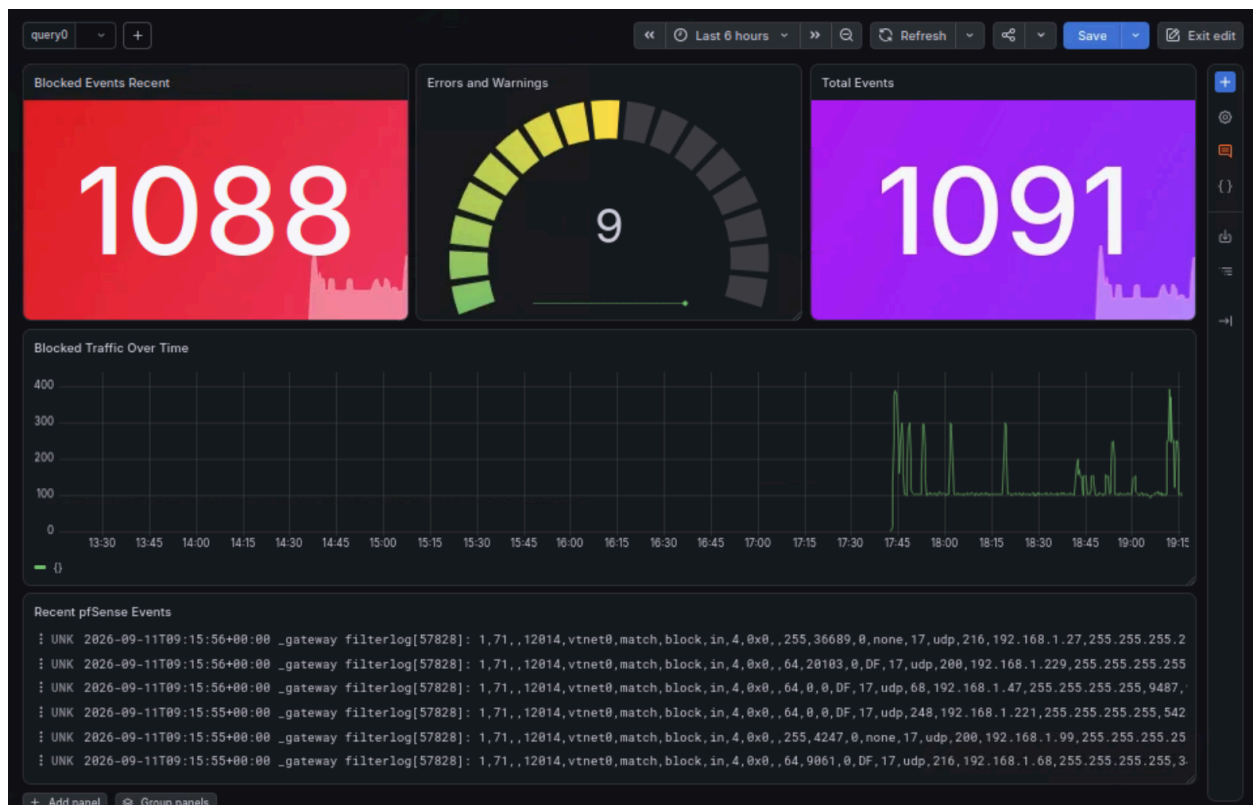


Figure 5: Grafana dashboard displaying network activity

Once Loki was connected to Grafana, I created a custom dashboard to make the pfSense logs easier to view and understand. Using queries, I was able to separate the different types of events and display them in panels.



Dashboard Queries	
Panel	Query code
Blocked Events Recent	sum(count_over_time({job="pfsense"} != ";block," [\$__range]))
Total Events	sum(count_over_time({job="pfsense"} [\$__range]))
Recent pfSense Events	{job="pfsense"}
Blocked Traffic Over Time	sum(count_over_time({job="pfsense"} != ";block," [\$__interval]))
Errors and Warnings	sum(count_over_time({job="pfsense"} =~ "(?i)(error warning failed)" [\$__range]))

Monitoring Access and Security

The Grafana dashboard is being hosted from the HQ-MONITOR through the address 192.168.90.60 using TCP port 3000. To restrict access to only staff with administrator permissions, I configured pfSense so only the HQ-MGMT-PC on VLAN 99 can connect to the dashboard. This keeps the dashboard unavailable to general staff on Employee and Visitor networks.

Rules (Drag to Change Order)											
	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓ 2/76 KIB	IPv4 TCP	MANAGEMENT_VLAN subnets	*	This Firewall (self)	MANAGEMENT_PORTS	*	none		R10b -	
☐	✓ 0/0 B	IPv4 TCP	MANAGEMENT_VLAN subnets	*	This Firewall (self)	MANAGEMENT_PORTS	*	none		R10a - Allow Management Access to pfSense	
☐	✓ 0/16 KIB	IPv4 TCP/UDP	HQ_AP	*	*	DNS_PORT	*	none		HQ AP DNS	
☐	✓ 0/2.00 MIB	IPv4 TCP	HQ_AP	*	*	WEB_PORTS	*	none			
☐	✓ 0/0 B	IPv4 UDP	*	*	*	NTP_PORT	*	none			
☐	✓ 0/0 B	IPv4 TCP	192.168.99.71	*	192.168.90.60	3000 (HBCI)	*	none			

Figure 6: pfSense Management VLAN firewall rule allowing HQ-MGMT-PC on VLAN 99 to access the Grafana dashboard through port 3000.

Testing and Validation

Test	Expected Result	Result
SPAN capture	HQ-MONITOR should be able to see copied traffic from the different VLANs	PASS
Log storage	Received pfSense events should be saved into /var/log/pfsense.log	PASS
pfSense syslog	pfSense events should be received by HQ-MONITOR	PASS
Alloy to Loki	Alloy should read the pfSense log file and send the events into Loki	PASS
Grafana dashboard	pfSense events should be visible through the Grafana dashboard	PASS
Management access	HQ-MGMT-PC should be able to open the Grafana dashboard on port 3000	PASS

Security, Risk and Change Management

To provide security access to the Grafana dashboard was limited to the HQ-MGMT-PC using a pfSense firewall rule. This ensures Employee and Visitor devices are prevented from accessing the monitoring system and reduces the risk of the dashboard configuration being changed or viewed by unauthorised people connected to the network. When improving the logging and monitoring solution in the Brisbane HQ, I kept the original rsyslog configuration working and added Grafana, Loki and Alloy as an addition to it. Allowing improvements without interrupting the existing solution.

Final Implementation

The monitoring and logging for the Brisbane HQ was implemented and tested without issue, which provides an intuitive way to view network and security traffic that aligns with our project MVP.